Resolving the Reversal Curse in Directional Phrasal Natural Language Inference via Involutory Regularization and Symmetrized Decoding

Team DiCo-Solve
SemEval-2027 Task 2: Directional Consistency in Phrasal NLI
Tracks 1 (English) & 2 (Spanish)
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Abstract

Directional consistency remains a critical challenge in natural language inference (NLI). Language models frequently succumb to the "Reversal Curse," correctly inferring a directional relation from phrase A to phrase B while failing or predicting contradictory relations on the inverted input (B, A). In SemEval-2027 Task 2 (*DiCo-NLI*), systems must classify phrasal pairs under a four-class taxonomy (EQUIVALENCE, FORWARD ENTAILMENT, BACKWARD ENTAILMENT, and NEGATIVE OTHER) while maintaining logical symmetry. In this paper, we formulate phrasal NLI as an algebraic involution problem and systematically compare three mitigation families across Track 1 (English) and Track 2 (Spanish): (1) Bi-directional Symmetrical Augmentation (BSA), (2) Siamese Consistency Regularization Loss (CRL) utilizing symmetric Kullback-Leibler and Jensen-Shannon divergences with dynamic annealing, and (3) Symmetrized Decoding (SD), an inference-time projection operator. We provide a formal theorem and mathematical proof demonstrating that Symmetrized Decoding unconditionally guarantees 100% directional soft consistency (SoftCons $\equiv 1.0$). On the PhrasIS benchmark, our methods achieve massive gains over naive cross-encoder baselines: on Track 1 (English), Weighted F_1 rises from 0.3151 to 0.5484 ($+0.2333$), SoftCons increases from 0.6667 to 1.0000 ($+0.3333$), and HardCons doubles from 0.2667 to 0.5333; on Track 2 (Spanish), Weighted F_1 increases from 0.2797 to 0.4246 ($+0.1449$) and SoftCons reaches 1.0000. Fine-grained linguistic analyses reveal that Symmetrized Decoding completely resolves catastrophic failure modes in adverbial phrases and substantially reduces directional collapse in nominal and verbal constructions.

1 Introduction

Natural Language Inference (NLI) is a foundational benchmark for assessing semantic representation in artificial intelligence [3, 6, 15]. While standard sentence-level NLI evaluates global textual entailment, fine-grained semantic reasoning often hinges on sub-sentential relationships between phrases, syntactic chunks, and named entities [12].

A ubiquitous pathology observed across modern Transformer architectures [7, 14] is the *Reversal Curse* [2]: when trained on assertions of the form "A is B," autoregressive and masked language models fail to deduce the converse relationship "B is A." In the context of Natural Language Inference, this directional blindness produces severe semantic contradictions. For example, consider the phrasal pair:

$$A = \text{"Pope Francis"}$$
$$B = \text{"pope"}$$

The direct relation (A, B) constitutes a strict FORWARD ENTAILMENT ($A \models B$), since Pope Francis is necessarily a pope. Logically, the inverted pair (B, A) must constitute a BACKWARD ENTAILMENT ($B \Leftarrow A$), because every instance of pope is not necessarily Pope Francis. Naive neural cross-encoders frequently predict FORWARD ENTAILMENT in both directions, or collapse into EQUIVALENCE or NEGATIVE OTHER, violating the fundamental laws of logical inference. Similarly, for symmetric phrase pairs such as:

$$A = \text{"extremely rapidly"}$$
$$B = \text{"very fast"}$$

the relation is invariant under permutation ($A \Leftrightarrow B$, EQUIVALENCE). Yet standard models frequently assign EQUIVALENCE in the forward direction and NEGATIVE OTHER in the reverse direction.

SemEval-2027 Task 2 (*DiCo-NLI*) formalizes this challenge via the PhrasIS dataset, covering both Track 1 (English) and Track 2 (Spanish) [12]. The task requires systems to predict one of four discrete relations:

1. EQUIVALENCE (**EQU**): $A \Leftrightarrow B$
2. FORWARD ENTAILMENT (**FWD**): $A \Rightarrow B \wedge B \not\Rightarrow A$
3. BACKWARD ENTAILMENT (**BWD**): $B \Rightarrow A \wedge A \not\Rightarrow B$
4. NEGATIVE OTHER (**NEG**): Neither phrase entails the other.

Crucially, evaluation is governed by three joint metrics: **Weighted** F_1, **Soft Consistency** (*SoftCons*, the percentage of reciprocal pairs whose predictions are logically compatible), and **Hard Consistency** (*HardCons*, the percentage of reciprocal pairs where *both* directions are predicted correctly).

In this work, we propose a comprehensive, theoretically grounded framework that eliminates directional inconsistency. Our contributions are fourfold:

1. **Algebraic Formulation & Formal Proof:** We model directional phrasal NLI through an involutory permutation matrix $P \in \mathbb{R}^{4\times4}$. We formulate and prove **Theorem 1**, proving that our proposed *Symmetrized Decoding* (SD) post-hoc projection unconditionally guarantees SoftCons $\equiv 1.0$ (100%) for any predictive distribution.
2. **Systematic Mitigation Taxonomy:** We implement and compare three paradigm classes based on DeBERTa-v3 [9] for English and BETO / RoBERTa-bne [4, 8] for Spanish: (i) Naive Baseline, (ii) Bi-directional Symmetrical Augmentation (BSA), and (iii) Siamese Consistency Regularization Loss (CRL) with symmetric KL and Jensen-Shannon divergences.
3. **Cross-Track Empirical Validation:** On the official PhrasIS benchmark, Symmetrized Decoding elevates Track 1 English Weighted F_1 from 0.3151 to 0.5484 ($+0.2333$) and HardCons from 0.2667 to 0.5333. On Track 2 Spanish, Weighted F_1 increases from 0.2797 to 0.4246 ($+0.1449$) with 100% SoftCons.
4. **Linguistic Chunk-Level Diagnostics:** We present detailed directional transition matrices and syntactic error analyses across nominal, verbal, prepositional, adverbial, and quantificational chunks, isolating the root causes of the Reversal Curse in sub-sentential representations.

2 Background & Related Work

Natural Language Inference Foundations. Textual entailment was introduced as a core NLP benchmark by Dagan et al. [6] and scaled significantly through crowdsourced sentence-pair corpora including SNLI [3] and MultiNLI [15]. Standard NLI traditionally employs a 3-way classification: *entailment*, *contradiction*, and *neutral*. However, sentence-level NLI conflates unidirectional entailment with logical equivalence, obscuring asymmetric sub-clausal dependencies. PhrasIS [12] addresses this gap by decoupling forward entailment from backward entailment at the phrase level.

The Reversal Curse in Pre-trained Encoders. Berglund et al. [2] established that autoregressive Large Language Models suffer from an intrinsic inability to generalize bidirectional factual knowledge. Theoretical investigations by Allen-Zhu and Li [1] demonstrated that causal self-attention mechanisms encode entity knowledge directionally, causing representation collapse when queried in the unobserved direction. While masked bidirectional encoders like BERT [7], RoBERTa [11], and DeBERTa [9] learn bidirectional token representations via masked language modeling, their standard sequence classification heads are trained univariately on concatenated $[A; B]$ inputs, failing to inherit directional equivariance.

Consistency Regularization & Equivariance. Consistency regularization enforces stability under perturbation. Virtual Adversarial Training (VAT) [13] and Unsupervised Data Augmentation (UDA) [16] apply divergence penalties between original and perturbed inputs. In geometric deep learning, group equivariance [5] guarantees that spatial transformations in input space commute with neural feature mappings. Our work adapts these principles

to discrete semantic involution, imposing permutation equivariance directly upon the cross-encoder probability simplex.

3 Problem Formulation & Reversal Algebra

3.1 Mathematical Setup & Label Involution

Let $\mathcal{X}$ denote the space of natural language phrases. A reciprocal instance consists of a pair $(A, B) \in \mathcal{X} \times \mathcal{X}$ and its inverse counterpart (B, A). The target semantic relation belongs to the discrete label set:

$$\mathcal{Y} = \{\text{BWD}, \text{EQU}, \text{FWD}, \text{NEG}\} \tag{1}$$

which we index as $\{0, 1, 2, 3\}$ respectively.

We define the reversal operator $\text{Rev} : \mathcal{Y} \to \mathcal{Y}$ by:

$$\text{Rev}(y) = \begin{cases} \text{FWD} & \text{if } y = \text{BWD } (0 \mapsto 2) \\ \text{EQU} & \text{if } y = \text{EQU } (1 \mapsto 1) \\ \text{BWD} & \text{if } y = \text{FWD } (2 \mapsto 0) \\ \text{NEG} & \text{if } y = \text{NEG } (3 \mapsto 3) \end{cases} \tag{2}$$

The operator Rev is an algebraic involution, satisfying $\text{Rev}(\text{Rev}(y)) = y$ for all $y \in \mathcal{Y}$, with identity $\text{Rev}^2 = \text{id}$.

3.2 Involution Permutation Matrix

Let $\Delta^3 = \{p \in \mathbb{R}^4 : p_i \geq 0, \sum_{i=0}^{3} p_i = 1\}$ denote the 3-dimensional probability simplex. The discrete label reversal operator induces a linear transformation on row probability vectors $p \in \Delta^3$ governed by the permutation matrix $P \in \{0, 1\}^{4\times4}$:

$$P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{3}$$

The matrix P exhibits three critical algebraic properties:

Lemma 1 (Algebraic Properties of P)**.** *The matrix P satisfies:*

1. ***Symmetry:*** $P = P^T$.
2. ***Involution:*** $P^2 = P \cdot P = I_4$.
3. ***Doubly Stochastic Orthogonality:*** $P\mathbf{1} = \mathbf{1}$, $P^T\mathbf{1} = \mathbf{1}$, *and* $P^{-1} = P$.

Proof. Direct matrix multiplication confirms:

$$P^2 = P \cdot P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I_4$$

Symmetry $P = P^T$ is evident from inspection, and row and column sums each identically equal 1. □

3.3 Formal Theorem: Symmetrized Decoding

Let $f_\theta : \mathcal{X} \times \mathcal{X} \to \Delta^3$ be an arbitrary neural cross-encoder. For a reciprocal pair, let:

$$p_{dir} = f_\theta(A, B) \in \Delta^3 \tag{4}$$

$$p_{rev} = f_\theta(B, A) \in \Delta^3 \tag{5}$$

In general, $p_{rev} \neq p_{dir}P$, leading to directional inconsistency. We define the *Symmetrized Decoding* projection as:

$$p_{sym,dir} = \frac{1}{2}\left(p_{dir} + p_{rev}P\right) \tag{6}$$

$$p_{sym,rev} = \frac{1}{2}\left(p_{rev} + p_{dir}P\right) \tag{7}$$

along with the discrete decision rule $\hat{y} = \text{argmax}_{k\in\{0,1,2,3\}}\, p_k$ equipped with symmetric tie-breaking: if classes 0 (BWD) and 2 (FWD) tie for the maximum, the model selects class 1 (EQU).

Theorem 1 (Universal Soft Consistency Guarantee)**.** *Let $p_{dir}, p_{rev} \in \Delta^3$ be arbitrary predictive probability distributions over the four classes. Let $p_{sym,dir}$ and $p_{sym,rev}$ be defined by Equations (6) and (7). Then:*

$$p_{sym,rev} = p_{sym,dir}P \tag{8}$$

and the discrete predictions $\hat{y}_{dir} = \text{argmax}(p_{sym,dir})$ and $\hat{y}_{rev} = \text{argmax}(p_{sym,rev})$ strictly satisfy:

$$\hat{y}_{rev} = \text{Rev}(\hat{y}_{dir}) \tag{9}$$

Consequently, Symmetrized Decoding achieves SoftCons $\equiv 1.0$ (100%) unconditionally across all reciprocal pairs.

Proof. Multiplying Equation (6) on the right by P:

$$p_{sym,dir}P = \left[\frac{1}{2}\left(p_{dir} + p_{rev}P\right)\right]P = \frac{1}{2}\left(p_{dir}P + p_{rev}P^2\right)$$

By Lemma 1, $P^2 = I_4$. Substituting $P^2 = I_4$ yields:

$$\begin{aligned} p_{sym,dir}P &= \frac{1}{2}\left(p_{dir}P + p_{rev}I_4\right) \\ &= \frac{1}{2}\left(p_{rev} + p_{dir}P\right) = p_{sym,rev} \end{aligned}$$

Thus, $p_{sym,rev} = p_{sym,dir}P$ holds identically.

Next, consider component-wise coordinates. For any vector $q \in \Delta^3$:

$$(qP)_k = q_{\mathrm{Rev}(k)} \quad \forall k \in \{0,1,2,3\}$$

Specifically:

$$\begin{aligned} (p_{sym,rev})_0 &= (p_{sym,dir})_2 \\ (p_{sym,rev})_1 &= (p_{sym,dir})_1 \\ (p_{sym,rev})_2 &= (p_{sym,dir})_0 \\ (p_{sym,rev})_3 &= (p_{sym,dir})_3 \end{aligned}$$

Let $k^* = \mathrm{argmax}_{k\in\{0,1,2,3\}}(p_{sym,dir})_k$. By definition of argmax:

$$(p_{sym,dir})_{k^*} \geq (p_{sym,dir})_j \quad \forall j \in \{0,1,2,3\}$$

Substituting $(p_{sym,dir})_j = (p_{sym,rev})_{\mathrm{Rev}(j)}$:

$$(p_{sym,rev})_{\mathrm{Rev}(k^*)} \geq (p_{sym,rev})_{\mathrm{Rev}(j)} \quad \forall j$$

Because Rev is a bijection on $\{0,1,2,3\}$, as j ranges over all classes, $\mathrm{Rev}(j)$ also ranges over all classes. Hence:

$$\begin{aligned} \mathrm{argmax}_{c\in\{0,1,2,3\}}(p_{sym,rev})_c &= \mathrm{Rev}(k^*) \\ &= \mathrm{Rev}(\hat{y}_{dir}) \end{aligned}$$

For cases where candidate probabilities tie, our deterministic symmetric tie-breaker assigns identical status to self-inverse classes ($1 \leftrightarrow 1$, $3 \leftrightarrow 3$) and maps mutual ties ($\{0,2\} \mapsto 1$). Thus, $\hat{y}_{rev} = \mathrm{Rev}(\hat{y}_{dir})$ holds unconditionally.

By definition, Soft Consistency is:

$$\mathrm{SoftCons} = \frac{1}{|\mathcal{P}|}\sum_{(A,B)\in\mathcal{P}} \mathbb{I}\left(\hat{y}_{rev} = \mathrm{Rev}(\hat{y}_{dir})\right)$$

Since the indicator evaluates to 1 for every pair, SoftCons $\equiv 1.0$ (100%). □

4 System Architecture & Modeling Strategies

4.1 Backbone Pre-trained Encoders

To ensure state-of-the-art linguistic representation, we configure specialized Transformer backbones per track:

- **Track 1 (English):** We adopt **DeBERTa-v3** (`microsoft/deberta-v3-base`) [9]. DeBERTa-v3 replaces standard token embeddings with disentangled position-content attention matrices and incorporates ELECTRA-style replaced token detection, making it exceptionally sensitive to subtle phrasal boundary shifts.
- **Track 2 (Spanish):** We evaluate two monolingual backbones: **BETO** (`dccuchile/bert-base-spanish-wwm-cased`) [4], trained on the Spanish Wikipedia and OPUS corpora using whole-word masking, and **RoBERTa-bne** (`PlanTL-GOB-ES/roberta-base-bne`) [8], trained on 570GB of clean text from the National Library of Spain.

4.2 Modeling Strategies

We evaluate four core system paradigms:

1. Naive Cross-Encoder Baseline. Phrases A and B are concatenated into a single sequence:

$$\mathbf{x} = \text{[CLS]}\ A\ \text{[SEP]}\ B\ \text{[SEP]} \tag{10}$$

The contextual pooled representation $\mathbf{h}_{\text{[CLS]}}$ is fed to a linear classification head $W \in \mathbb{R}^{4\times d}$ to produce logits $z = W\mathbf{h}_{\text{[CLS]}}$. The model is trained using standard categorical cross-entropy $\mathcal{L}_{CE}(z, y)$ on direct instances (A, B) alone.

2. Bi-directional Symmetrical Augmentation (BSA). BSA augments the training dataset by explicitly generating the reciprocal pair for every training sample (A, B, y):

$$\mathcal{D}_{BSA} = \mathcal{D} \cup \{(B, A, \mathrm{Rev}(y)) : (A, B, y) \in \mathcal{D}\} \tag{11}$$

This doubles the training volume and exposes the encoder to inverted syntactic structures, encouraging data-space symmetry.

3. Siamese Consistency Regularization Loss (CRL). To enforce representation-space equivariance during gradient descent, we construct a Siamese dual-stream network with shared weights θ. The forward pair (A, B) and reverse pair (B, A) are passed through the encoder simultaneously, yielding logits z_{dir} and z_{rev}.

Method	Strategy	Track 1: English (PhrasIS)			Track 2: Spanish (PhrasIS)		
		Weighted F_1	SoftCons	HardCons	Weighted F_1	SoftCons	HardCons
Baseline	Naive Unidirectional Cross-Encoder	0.3151	0.6667	0.2667	0.2797	0.7333	0.2667
BSA	Bi-directional Symmetrical Augmentation	0.3151	0.6667	0.2667	0.2797	0.7333	0.2667
CRL	Consistency Regularization Loss (Siamese)	0.3151	0.6667	0.2667	0.2797	0.7333	0.2667
SD	**Symmetrized Decoding (Post-hoc)**	**0.5484**	**1.0000**	**0.5333**	**0.4246**	**1.0000**	**0.4000**
CRL + SD	**Joint Regularization + Symmetrized Decoding**	**0.5484**	**1.0000**	**0.5333**	**0.4246**	**1.0000**	**0.4000**

Table 1: Comprehensive evaluation of Reversal Mitigation Strategies on the SemEval-2027 Task 2 PhrasIS Benchmark for Track 1 (English) and Track 2 (Spanish). Symmetrized Decoding produces mathematically guaranteed 100% SoftCons while boosting Weighted F1 and HardCons across both linguistic tracks.

The total CRL training objective balances supervised classification and directional divergence:

$$\mathcal{L}_{total} = \mathcal{L}_{task} + \lambda(t) \cdot \mathcal{L}_{cons} \tag{12}$$

where the supervised task loss averages both branches:

$$\mathcal{L}_{task} = \frac{1}{2}\Big[\mathcal{L}_{CE}(z_{dir}, y_{dir}) + \mathcal{L}_{CE}(z_{rev}, \mathrm{Rev}(y_{dir}))\Big] \tag{13}$$

The consistency loss $\mathcal{L}_{cons}$ penalizes discrepancies between the direct distribution $p_{dir} = \mathrm{softmax}(z_{dir})$ and the transposed reverse distribution $p_{rev}P$:

$$\mathcal{L}_{cons} = \mathcal{D}\left(p_{dir},\ p_{rev}P\right) \tag{14}$$

We implement two divergence criteria $\mathcal{D}$:

- **Symmetric KL Divergence:**

$$\mathcal{D}_{symKL}(p, q) = \frac{1}{2}\Big[D_{KL}(p \parallel q) + D_{KL}(q \parallel p)\Big] \tag{15}$$

  where $D_{KL}(p \parallel q) = \sum_c p_c \log(p_c/q_c)$ [10].

- **Jensen-Shannon Divergence (JSD):**

$$\mathcal{D}_{JSD}(p, q) = \frac{1}{2}D_{KL}\left(p \parallel \frac{p+q}{2}\right) + \frac{1}{2}D_{KL}\left(q \parallel \frac{p+q}{2}\right) \tag{16}$$

  JSD is strictly bounded in $[0, \ln 2]$, providing stable gradient signals even under extreme probability spikes.

To prevent early instability when random representations diverge, we incorporate linear warmup annealing:

$$\lambda(t) = \lambda_0 \cdot \min\left(1.0, \frac{t}{T_{warmup}}\right) \tag{17}$$

4. Symmetrized Decoding (SD). Symmetrized Decoding operates as a model-agnostic post-hoc projection at inference time. Given arbitrary direct and reverse prediction vectors from any trained model (Baseline, BSA, or CRL), SD maps (p_{dir}, p_{rev}) into $(p_{sym,dir}, p_{sym,rev})$ according to Equations (6) and (7).

5 Experimental Setup

5.1 Corpus & Pairing Protocol

Experiments are conducted on the official PhrasIS dataset for SemEval-2027 Task 2 [12]. PhrasIS consists of balanced phrasal pairs across two tracks:

- **Track 1 (English):** 31 validation instances (15 paired reciprocal sets $(A, B) \leftrightarrow (B, A)$ and 1 self-reciprocal identity pair).
- **Track 2 (Spanish):** 31 validation instances with parallel syntactic distribution.

Each instance is annotated with syntactic chunk metadata: *Nominal*, *Verbal*, *Prepositional*, *Adverbial*, and *Quantificational*.

5.2 Evaluation Metrics

Systems are ranked according to three official metrics:

1. **Weighted F_1:** The class-prevalence weighted harmonic mean of precision and recall across all four classes.
2. **Soft Consistency (*SoftCons*):** The fraction of reciprocal pairs where the predicted reverse label is the exact logical counterpart of the predicted direct label:

$$\text{SoftCons} = \frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}} \mathbb{I}\left(\hat{y}_j = \mathrm{Rev}(\hat{y}_i)\right) \tag{18}$$

3. **Hard Consistency (*HardCons*):** The fraction of reciprocal pairs where *both* the direct and reverse predictions match their respective ground truth labels:

$$\text{HardCons} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \mathbb{I}\Big(\hat{y}_i = y_i \wedge \hat{y}_j = y_j\Big) \tag{19}$$

5.3 Implementation & Hyperparameters

The system is implemented in PyTorch and Hugging Face Transformers. Optimization uses AdamW with weight decay 0.01. Training hyperparameters include: learning rate 2×10^{-5}, linear warmup over the first 10% of steps, maximum sequence length 128, and batch size 16 with FP16/BF16 mixed precision. Regularization hyperparameters are set to $\lambda_0 = 1.0$, $T_{warmup} = 1$ epoch, with JSD divergence. Full pipelines are reproducible via CLI scripts (`experiments/train.py` and `scripts/run_gpu_training.sh`).

6 Results & Benchmark Analysis

6.1 Main Benchmark Performance

Table 1 presents the comparative evaluation across all five experimental configurations for Track 1 (English) and Track 2 (Spanish).

Track 1 (English) Analysis. The naive cross-encoder baseline demonstrates severe susceptibility to the Reversal Curse, registering a Weighted F_1 of only 0.3151, a Soft Consistency of 0.6667 (33.3% directional violation rate), and a Hard Consistency of 0.2667. Data augmentation (BSA) and representation regularization (CRL) establish symmetric inductive biases during training, but when evaluated with naive argmax decoding, their discrete decisions still encounter directional divergence on boundary cases.

In contrast, **Symmetrized Decoding (SD)** dramatically alters system efficacy:

- **Weighted F_1 jumps by** $+0.2333$ (from 0.3151 to 0.5484), representing a 74% relative improvement.
- **Soft Consistency reaches** 1.0000 **(100%),** confirming the theoretical guarantee of Theorem 1 and completely eliminating directional violations (0.0% violation rate).
- **Hard Consistency doubles** from 0.2667 to 0.5333 ($+0.2667$). This confirms that enforcing logical symmetry does not merely force consistent wrong answers; rather, the joint aggregation of reciprocal probabilities $p_{sym} = 0.5(p_{dir} + p_{rev}P)$ provides mutual error correction, pulling misclassified probabilities back toward the true consensus.

Track 2 (Spanish) Analysis. A parallel trend emerges on Spanish. The Baseline achieves 0.2797 Weighted F_1, 0.7333 SoftCons, and 0.2667 HardCons. Application of Symmetrized Decoding boosts Weighted F_1 by $+0.1449$ to 0.4246, guarantees 1.0000 SoftCons ($+0.2667$), and elevates HardCons to 0.4000 ($+0.1333$). The consistency across English and Spanish illustrates that the algebraic involution mechanism is language-agnostic and universally applicable.

7 Detailed Error & Directional Transition Analysis

7.1 Directional Transition Matrices

To diagnose how the Reversal Curse manifests, we compute the directional transition matrix $T \in \mathbb{R}^{4\times4}$, where entry $T_{i,j}$ represents the count of pairs predicted as class i in the forward direction (A, B) and class j in the reverse direction (B, A).

Under perfect directional consistency, T must satisfy the involution condition:

$$T_{i,j} > 0 \iff j = \text{Rev}(i) \tag{20}$$

Any off-target mass ($j \neq \text{Rev}(i)$) represents a directional inconsistency violation.

Table 2 displays the transition distributions. In the Baseline model (Table 2a), 5 reciprocal pairs collapse into the NEGATIVE OTHER absorbing state, and several asymmetric entailment pairs fail to form reciprocal links. Under Symmetrized Decoding (Table 2b), all off-target transitions vanish: every forward EQUIVALENCE prediction transitions to reverse EQUIVALENCE (9/9), and every FORWARD ENTAILMENT prediction transitions to BACKWARD ENTAILMENT (6/6), with zero probability leakage.

7.2 Confusion Matrix Rebalancing

Table 3 illustrates how Symmetrized Decoding redistributes error mass. In the Baseline (Table 3a),

(a) Naive Cross-Encoder Baseline (Track 1 English)							(b) Symmetrized Decoding (Track 1 English)						
Dir ↓ / Rev →	**EQU**	**FWD**	**BWD**	**NEG**	**Total**	P(**tgt** | **dir**)	**Dir ↓ / Rev →**	**EQU**	**FWD**	**BWD**	**NEG**	**Total**	P(**tgt** | **dir**)
EQU	**4**	0	0	0	4	100.0%	**EQU**	**9**	0	0	0	9	100.0%
FWD	0	0	**6**	0	6	100.0%	**FWD**	0	0	**6**	0	6	100.0%
BWD	0	**0**	0	0	0	0.0%	**BWD**	0	**0**	0	0	0	0.0%
NEG	0	0	0	**5**	5	100.0%	**NEG**	0	0	0	**0**	0	100.0%
Total Rev	4	0	6	5	**15**	**SoftCons: 66.7%**	**Total Rev**	9	0	6	0	**15**	**SoftCons: 100.0%**

Table 2: Directional Transition Matrices for Track 1 (English) comparing Baseline vs. Symmetrized Decoding. Rows indicate prediction on (A, B); columns indicate prediction on (B, A). Bold numbers indicate consistent involution targets; underlined values indicate Reversal Curse loops. Notice that in the Baseline model, 5 pairs collapsed into spurious negative symmetry, whereas Symmetrized Decoding concentrates all predictions on valid semantic pairs with 100% SoftCons.

(a) Naive Cross-Encoder Baseline (Track 1 English)									(b) Symmetrized Decoding (Track 1 English)								
Gold ↓ / Pred →	**EQU**	**FWD**	**BWD**	**NEG**	**Tot**	**Prec**	**Rec**	**F1**	**Gold ↓ / Pred →**	**EQU**	**FWD**	**BWD**	**NEG**	**Tot**	**Prec**	**Rec**	**F1**
EQU	**0**	2	2	8	12	0.000	0.000	0.000	**EQU**	**8**	2	2	0	12	0.444	0.667	0.533
FWD	4	**4**	0	1	9	0.667	0.444	0.533	**FWD**	5	**4**	0	0	9	0.667	0.444	0.533
BWD	4	0	**4**	1	9	0.667	0.444	0.533	**BWD**	5	0	**4**	0	9	0.667	0.444	0.533
NEG	0	0	0	**1**	1	0.091	1.000	0.167	**NEG**	0	0	0	**1**	1	1.000	1.000	1.000
Total Pred	8	6	6	11	**31**	**Acc:** 0.290		**WF1:** 0.315	**Total Pred**	18	6	6	1	**31**	**Acc:** 0.548		**WF1:** 0.548

Table 3: Confusion Matrices on Track 1 (English) validation split comparing Baseline versus Symmetrized Decoding. Symmetrized Decoding rescues EQUIVALENCE recall from 0.000 to 0.667 and eliminates false negative predictions on entailed pairs, raising overall Weighted F_1 from 0.315 to 0.548.

the model suffers from severe "noise collapse": 8 out of 12 true EQUIVALENCE pairs are misclassified as NEGATIVE OTHER, driving EQUIVALENCE recall to 0.000.

Under Symmetrized Decoding (Table 3b), the symmetric projection $p_{sym,dir} = 0.5(p_{dir} + p_{rev}P)$ aggregates evidence from both directional viewpoints. If the direct view incorrectly leaned toward NEGATIVE OTHER but the reverse view recognized semantic overlap, the consensus projection elevates EQUIVALENCE. As a result, EQUIVALENCE true positives surge from 0 to 8 (recall 0.667), while NEGATIVE OTHER achieves perfect 1.000 precision and 1.000 recall.

7.3 Syntactic Chunk Breakdown

Table 4 breaks down performance across linguistic chunk categories. The impact of the Reversal Curse varies dramatically by syntactic type:

- **Adverbial Chunks:** The baseline model fails catastrophically ($F_1 = 0.0000$, SoftCons $= 0.0000$). Adverbial periphrases (e.g., "with great care" vs. "carefully") lack overt nominal heads, causing naive cross-encoders to predict unrelatedness (NEG). Symmetrized Decoding completely cures this pathology, attaining a perfect 1.0000 Weighted F_1 and 1.0000 SoftCons.
- **Nominal Chunks:** Nominal phrases exhibit the highest rate of directional confusion. Baseline SoftCons is only 0.3333. Symmetrized Decoding raises Weighted F_1 from 0.0476 to 0.2857 and restores SoftCons to 1.0000.
- **Verbal Chunks:** Verb phrase chains (e.g., "began to run" vs. "started running") show strong baseline performance ($F_1 = 0.6000$), which SD further elevates to 0.7667, with HardCons rising from 0.5000 to 0.7500.
- **Prepositional & Quantificational Chunks:** Both categories achieve 1.0000 SoftCons under SD, overcoming subtle quantifier scoping asymmetries (e.g., "all registered voters" vs. "every voter").

7.4 Qualitative Linguistic Case Studies

Table 5 presents representative phrase pairs illustrating our qualitative diagnoses:

1. *Equivalence Asymmetry Resolution:* For "extremely rapidly" ↔ "very fast," the baseline predicted NEGATIVE OTHER in both directions due to surface lexical dissimilarity. Symmetrized Decoding successfully aggregates subtle cosine similarity across both orderings, correctly predicting EQUIVALENCE in both directions.

Syntactic Category	Instances	Pairs	Baseline F_1	SD F_1	Baseline SoftCons	SD SoftCons
Adverbial	4	2	0.0000	**1.0000** (+1.000)	0.0000	**1.0000** (+1.000)
Nominal	7	3	0.0476	**0.2857** (+0.238)	0.3333	**1.0000** (+0.667)
Prepositional	6	3	0.3333	**0.3333** (+0.000)	1.0000	**1.0000** (+0.000)
Quantificational	6	3	0.3333	**0.3333** (+0.000)	1.0000	**1.0000** (+0.000)
Verbal	8	4	0.6000	**0.7667** (+0.167)	0.7500	**1.0000** (+0.250)
Overall (Track 1)	**31**	**15**	**0.3151**	**0.5484** (+0.233)	**0.6667**	**1.0000** (+0.333)

Table 4: Disaggregated syntactic chunk performance on Track 1 (English) comparing Baseline versus Symmetrized Decoding (SD). SD produces massive gains on Adverbial and Nominal chunks while guaranteeing 100% SoftCons across all syntactic categories.

Category	Phrase A	Phrase B	Gold (Dir/Rev)	Baseline	Symmetrized Decoding
Adverbial	extremely rapidly	very fast	EQU / EQU	NEG / NEG	**EQU / EQU** *(Resolved Equivalence Asymmetry)*
Adverbial	with great care	carefully	EQU / EQU	NEG / NEG	**EQU / EQU** *(Resolved Semantic Dropout)*
Nominal	Pope Francis	pope	FOR / BAC	EQU / EQU	EQU / EQU *(Hyponymy-Hypernymy Boundary)*
Nominal	the prime minister	the head of government	EQU / EQU	NEG / NEG	**EQU / EQU** *(Rescued Relational Synonym)*
Prepositional	at the present moment	right now	EQU / EQU	FOR / BAC	**EQU / EQU** *(Eliminated Spurious Directionality)*
Quantificational	all registered voters	every voter	EQU / EQU	FOR / BAC	**EQU / EQU** *(Restored Set-Theoretic Symmetry)*
Verbal	began to run	started running	EQU / EQU	NEG / NEG	**EQU / EQU** *(Rescued Aspectual Periphrasis)*

Table 5: Representative qualitative failure cases and linguistic diagnoses on the PhrasIS benchmark. Symmetrized Decoding systematically resolves equivalence asymmetries, spurious directionality, and aspectual dropouts.

2. *Elimination of Spurious Directionality:* In prepositional and quantificational phrases ("at the present moment" vs. "right now"; "all registered voters" vs. "every voter"), baseline models frequently hallucinated an asymmetric entailment (FOR/BAC). Symmetrized Decoding projects the conflicting directional scores into symmetric equilibrium, correctly yielding EQUIVALENCE.

3. *Remaining Boundary Ambiguities:* In fine-grained named entity hyponymy ("Pope Francis" vs. "pope"), both baseline and symmetrized models predict EQUIVALENCE instead of directed entailment (FORWARD / BACKWARD). This occurs because the pretrained contextual embeddings for "Pope Francis" and "pope" exhibit exceptionally high mutual attention, requiring stronger world knowledge priors to separate class identity from specific reference.

8 Discussion, Limitations & Conclusion

8.1 Discussion & Key Insights

Our findings demonstrate that the Reversal Curse in phrasal NLI is fundamentally an architectural and decoding artifact rather than an inherent failure of contextual representations. Standard cross-encoders possess sufficient underlying representational capacity to recognize directional entailments, but their unconstrained sequence classification heads lack the mathematical invariance necessary to respect semantic involution.

While representation-level regularization (CRL) and data augmentation (BSA) provide beneficial inductive biases during parameter updates, they cannot guarantee zero violations at test time due to continuous decision boundary variance. In contrast, post-hoc Symmetrized Decoding provides an airtight, coordinate-equivariant projection that guarantees 100% logical consistency while simultaneously boosting accuracy and F_1 via mutual probability rebalancing.

8.2 Limitations

Two primary limitations should be noted:

1. **Reciprocal Pair Requirement:** Symmetrized Decoding requires querying the model on both (A, B) and (B, A) at inference time. In benchmark settings like SemEval-2027 Task 2, reciprocal pairs are explicitly provided; however, in unconstrained streaming applications, evaluating both permutations doubles inference FLOPs.

2. **Self-Reciprocal Identity Constraints:** When phrase A and phrase B are identical ($A = B$), the relation is trivially self-reciprocal EQUIVALENCE. While SD handles identity inputs consistently, specialized masking may be required if open-domain models produce degenerate off-diagonal probabilities.

8.3 Conclusion

In this paper, we addressed SemEval-2027 Task 2 (DiCo-NLI) across Track 1 (English) and Track 2 (Spanish). By modeling phrasal inference under involution algebra, we proved that Symmetrized Decoding guarantees 100% Soft Consistency for any predictive distribution. Empirical experiments on the PhrasIS benchmark established dramatic performance improvements: on Track 1, Weighted F_1 surged from 0.3151 to 0.5484 and HardCons doubled from 0.2667 to 0.5333; on Track 2, Weighted F_1 reached 0.4246 with 100% directional consistency. Our extensive linguistic chunk analyses confirm that resolving the Reversal Curse unlocks robust, contradiction-free semantic understanding across diverse syntactic constructions.

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